INCH-POUND

MIL-PRF-32232
15 November 2006

PERFORMANCE SPECIFICATION

UNIVERSAL JOINT, RIGHT ANGLE, OFFSET, SHIPBOARD USE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers Right Angle Offset (RAO) universal joints designed to transmit up to 18,000 inch-pounds of torque between misaligned and/or offset shafts. RAO universal joints are double universal joints and can accommodate compound angles up to 90 degrees (45 degrees per joint in any direction) replacing pin-and-cube universal joints and gear boxes in remote operator arrangements installed on Navy surface ships and submarines.

1.2 Classification.

1.2.1 Types. RAO universal joints are of the following types (see Figure 1):

Type I - Two solid male end connectors

Type II - Two female end connections with slots to accommodate expansion and contraction

Type III - One solid male end and one female end connection

Type IV - Two male diameters with external keyways

Type V - Customer-defined end connection (see 3.4)

1.2.2 Sizes. RAO universal joint sizes will accommodate rigid input/output shafts with the following diameters:

Size (inches)	Size (inches)
⅜	⅞
½	1
⅝	1¼
¾	1½

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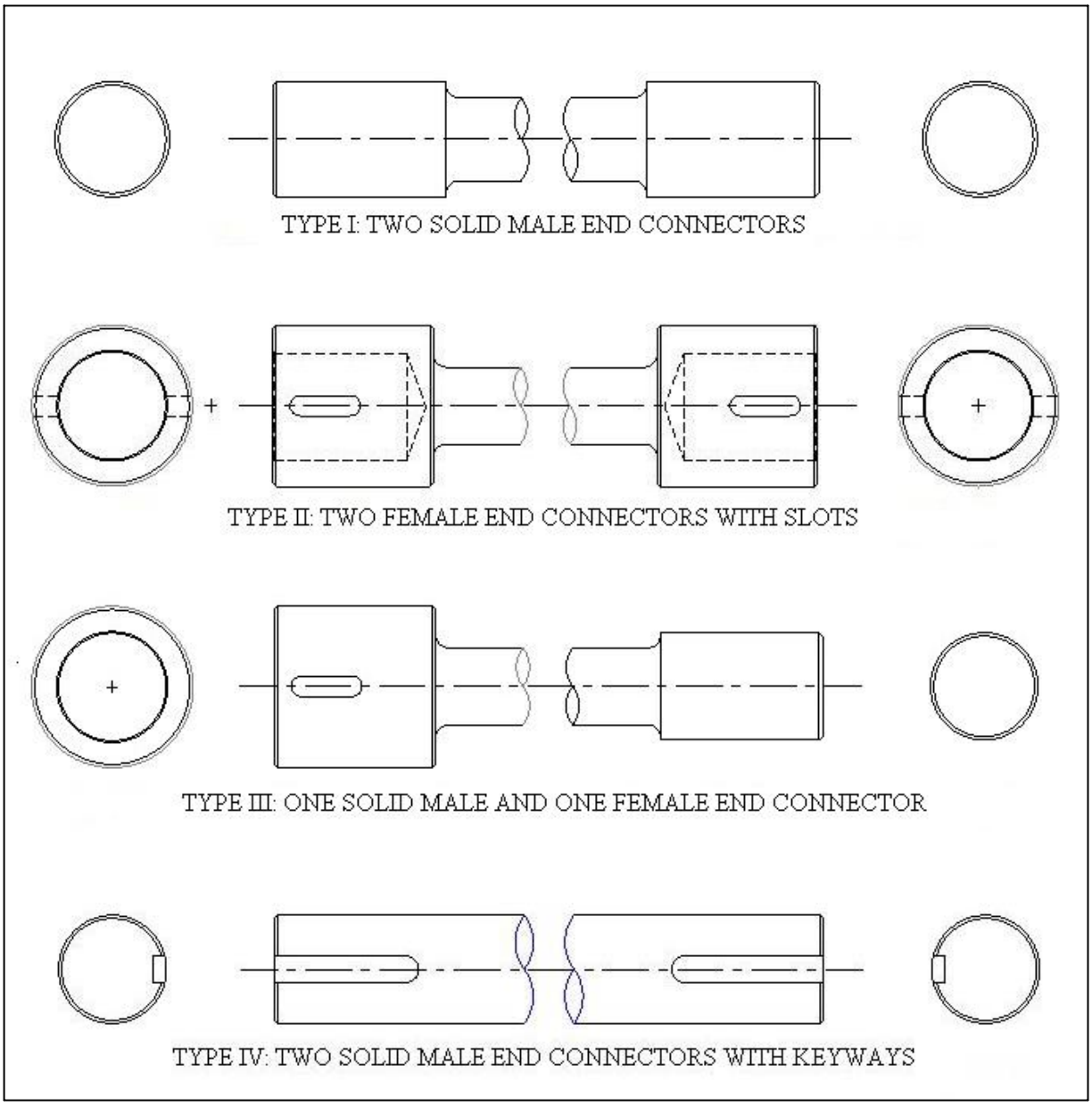


FIGURE 1. RAO universal joint end connections.

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1.3 Part or identifying number (PIN). PINs to be used for RAO universal joints acquired to this specification are created as follows:

M	**32232**	**-**	**X**	**X**
Prefix for Military Specification	Specification Number		Type (see code below)	Size (see code below)

Type	Type PIN Code
I	1
II	2
III	3
IV	4
V	5

Size (Inches)	Size PIN Code	Size (Inches)	Size PIN Code
⅜	A	1	F
½	B	1¼	G
⅝	C	1½	H
¾	D		
⅞	E		

Example: PIN M32232-1A is a Type I, ⅜-inch RAO universal joint.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-S-901 - Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment and Systems, Requirements for

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of this document are available from the Naval Logistics Library, 5450 Carlisle Pike, Mechanicsburg, PA 17055 or online at http://nll.ahf.nmci.navy.mil.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A106/A106M	-	Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service (DoD adopted)
ASTM B117	-	Standard Test Method for Salt Spray (Fog) Testing (DoD adopted)
ASTM D512	-	Standard Test Methods for Chloride Ion in Water (DoD adopted)
ASTM D1141	-	Specification for Substitute Ocean Water (DoD adopted)
ASTM D1179	-	Standard Test Methods for Fluoride Ion in Water (DoD adopted)
ASTM D1246	-	Standard Test Method for Bromide Ion in Water (DoD adopted)
ASTM D3223	-	Standard Test Method for Total Mercury in Water (DoD adopted)
ASTM D3557	-	Standard Test Methods for Cadmium in Water
ASTM D3559	-	Standard Test Methods for Lead in Water (DoD adopted)
ASTM D4190	-	Standard Test Method for Elements in Water by Direct-Current Argon Plasma Atomic Emission Spectroscopy
ASTM D4327	-	Standard Test Method for Anions in Water by Chemically Suppressed Ion Chromatography

(Copies of these documents are available from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or www.astm.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. RAO universal joints furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Materials. All components except bearings shall be made from passivated corrosion-resistant materials suitable for shipboard use and capable of meeting the required performance parameters specified herein. All materials shall be resistant to nitric acid. Materials selected shall maximize galvanic compatibility when installed in contact or close physical proximity to carbon steel, corrosion resisting steel, bronze, or copper-nickel alloys.

3.2.1 Detrimental materials. Detrimental materials shall not exceed the maximum concentrations as specified in Table I (see 4.5.10).

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TABLE I. Detrimental materials.

Material	Maximum concentration permitted (ppm) 1/
Asbestos	None permitted
Halogens (total of chloride, fluoride and bromide) 2/	1,000
Lead	250
Cadmium plating	None permitted
Mercury and compounds (none intentionally added) 3/	10
Sulfur	10,000

NOTES:

1/ ASTM Standards D512, D1141, D1179, D1246, and D4327 shall be used to ensure that the detrimental materials do not exceed the maximum permissible concentrations specified in Table I.

2/ Materials containing more than 1,000 parts per million (ppm) total halogens are acceptable provided only fluorine exceeds 250 ppm and total leachable halides do not exceed 250 ppm.

3/ During manufacture, fabrication, handling, packaging, and packing, RAO universal joint materials shall not come in contact with mercury or mercury compounds.

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Service life. RAO universal joints shall be designed for a minimum service life of 30 years and shall meet the endurance requirements of 3.6.7.

3.4 Design and construction. RAO universal joint design and construction may vary from types listed in 1.2.1 provided that the requirements of this document are met.

TABLE II. Axial and tensile loads.

Nominal size (inches)	Torque rating (in-lb)		Endurance torque (in-lb)	Torsional play (each joint)	
	Operating	Breakdown		Torque (in-lb)	Limit (degrees)
½	500	1500	250	4	0.625
⅝	500	1500	250	4	0.625
¾	2000	5000	1000	8	0.625
1	2000	5000	1000	8	0.625
1¼	3600	8750	1800	8	0.500
1½	18000	40000	9000	8	0.375

3.4.1 Angularity. RAO universal joints shall be capable of smooth near constant speed operation at angles up to 90 degrees, measured between the axes of the hubs (45 degrees per joint) (see Figure 2).

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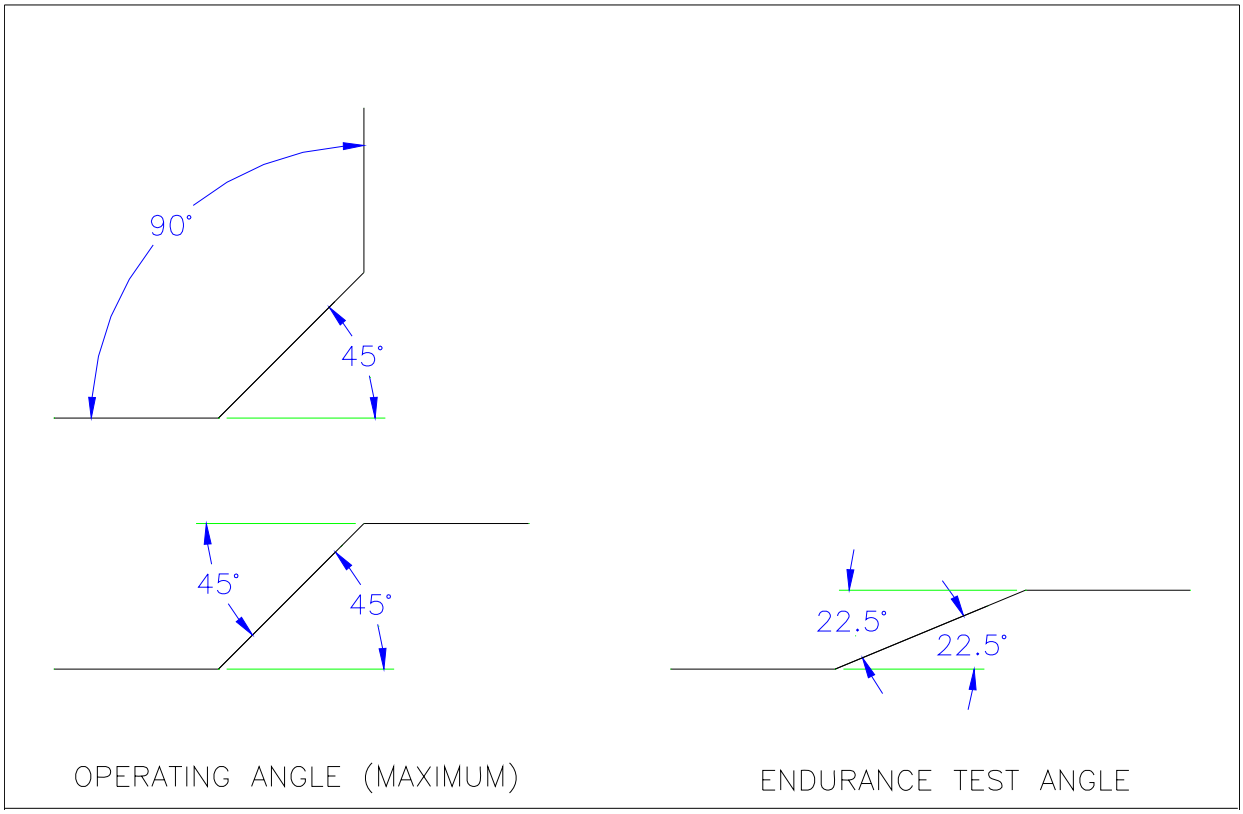


FIGURE 2. RAO angles.

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3.4.2 Lubrication. RAO universal joints shall utilize self-lubricating, dry bearings. RAO universal joints shall operate without added lubrication and shall not require lubrication covers or boots.

3.5 Physical requirements.

3.5.1 Surface roughness. RAO universal joint nonworking surfaces shall have a 250 micro-inch finish or better. RAO universal joint bearing surfaces shall have 32 micro-inch finish or better.

3.5.2 Marking. RAO universal joint marking requirements, if any, shall be as specified (see 6.2).

3.6 Performance requirements.

3.6.1 Operating environment. RAO universal joints shall perform in ambient temperatures ranging from -148 to 450 °F.

3.6.2 Salt spray. RAO universal joints shall withstand exposure to a salt spray marine environment when tested as specified in 4.5.2.

3.6.3 Torsional play. Torsional play shall not exceed the limit of Table II when the joint is subjected to the torque shown in Table II when tested as specified in 4.5.3.

3.6.4 Side play. The side play in RAO universal joints shall not exceed 0.015 inch when tested as specified in 4.5.4.

3.6.5 Tightness. The moment of force required to move each joint of the RAO universal joint through the angle of ±45 degrees shall not exceed 5.0 inch-pounds when subjected to the tightness test of 4.5.5.

3.6.6 Static torque. RAO universal joints shall withstand 100 percent of the operating torque loads shown in Table II without failure and must operate smoothly afterwards.

3.6.7 Endurance. At an angle of at least 22.5 degrees, RAO universal joints shall withstand continuous operation at 120 rpm for 12 hours under the endurance torque loads shown in Table II when configured as shown in Figure 2.

3.6.8 Shock resistance. RAO universal joints shall withstand Grade A, Class I, Type A shock test requirements of MIL-S-901.

3.6.9 Vibration resistance. RAO universal joints shall meet Type I requirements of MIL-STD-167-1.

3.7 Workmanship. Each RAO universal joint shall be uniform in quality and temper and shall be free from burrs, slivers, gouges, porosity, cracks or any other defects that may adversely affect serviceability. Since RAO universal joint components could be manufactured from castings, their surfaces may contain such defects. Parts made from castings shall be tested in accordance with T9074-AS-GIB-010/271 to verify that any defects will not adversely affect serviceability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be performed on one RAO universal joint of each size: ⅜-, ¾-, 1¼- and 1½-inch. This inspection shall include the examination of 4.4 and the tests of 4.5.2 through 4.5.10. Qualification by similarity and analysis for similarly constructed and successfully qualified items is acceptable.

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4.2.1 Notification of test failure. If during the performance of any portion of the qualification testing, any operating parameter or measurement is out of specification, or any abnormality is noted, the test shall stop, the reason and failure mode shall be noted and the cognizant technical point of contact shall be notified of the failure. Testing shall not proceed until the cognizant technical point of contact has evaluated the abnormality and given permission to resume testing.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4.

4.4 Examination. Each RAO universal joint shall be examined for compliance with the requirements specified in 3.2 through 3.5. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum performance shall constitute cause for rejection.

TABLE III. Examinations and inspections.

Test requirement	Requirement paragraph	Verification paragraph	Qualification	Conformance
Examination	3.2 through 3.5	4.4	X	X
Salt spray	3.6.2	4.5.2	X	
Torsional play	3.6.3	4.5.3	X	
Side play	3.6.4	4.5.4	X	
Tightness	3.6.5	4.5.5	X	
Static torque	3.6.6	4.5.6	X	
Endurance	3.6.7	4.5.7	X	
Shock resistance	3.6.8	4.5.8	X	
Vibration resistance	3.6.9	4.5.9	X	
Detrimental materials	3.6.10	4.5.10	X	

4.5 Methods of inspection.

4.5.1 Product certification. Material samples for RAO universal joints of each size (⅜-, ¾-, 1¼-, and 1½-inch) shall be tested as applicable to 4.5.10.

4.5.2 Salt spray. RAO universal joints shall be subjected to a salt spray test in accordance with ASTM B117. No appreciable corrosion or other damage shall be evident after exposure to the salt spray and the RAO universal joint shall operate smoothly through its entire range of motion. Each axis of the RAO universal joint shall operate freely when moved by hand through its full range of motion. In accordance with ASTM B117, the recommended exposure period, in increments of 24 hours, shall be mutually agreed upon by the manufacturer of the RAO universal joint and the procuring activity.

4.5.3 Torsional play. With one joint of the RAO fixed horizontally at zero degrees, an initial reading shall be taken with the torque applied to the RAO universal joint as specified in Table II. The torque shall be reversed to obtain the second reading. The torsional play at the circumference of the joint is the angular difference between the two readings and shall not exceed the limits shown in Table II. Repeat for the second element of the ROA universal joint. Torsional limits shall be determined after the endurance test.

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4.5.4 Side play. Side play of each axis of the RAO universal joint shall be determined by applying a 5-pound load in the sidewise direction to obtain an initial (zero) reading. The load is reversed to obtain a second reading. The total play is the difference between the two readings respectively for each axis. The maximum allowable side play shall be as specified in 3.6.4. Alternately, a feeler gage may be used to measure the maximum gap between side play elements.

4.5.5 Tightness. The moment of force required to move each end of the RAO universal joint about its horizontal axis through an angle of ±45 degrees shall be measured with the center section axis of the joint clamped in a vertical position. The moment of force shall not exceed the values shown in 3.6.5.

4.5.6 Static torque. Each joint of the RAO universal joint shall be configured at zero (0) degrees and shall be subjected to 90 percent of the operating torque specified in Table II without permanent deformation. Torque applied to RAO universal joints shall be increased to 100 percent. Joints shall not completely fail and shall operate smoothly after application of the full static torque load shown in Table II.

4.5.7 Endurance. At an angle of at least 22.5 degrees as shown in Figure 2, RAO universal joints shall be subjected to continuous operation at 120 rpm for 12 hours under the endurance torque loads specified in Table II. Endurance test equipment shall be of such design and construction that the test joint will be free of inadvertent end loads during the entire test period. Means shall also be provided to avoid high starting torque loads and the applied torque shall be substantially constant during the test. A smooth running mechanism free from vibration and shock loads shall be used. Forced cooling may be provided during the endurance test. At the completion of the test, torsional play shall be measured (see 4.5.3). Post endurance test joints shall exhibit less than 5 degrees of torsional play for each joint.

4.5.8 Shock resistance. RAO universal joints installed between valves and actuators (gear box, hydraulic or electric) shall be tested in accordance with Grade A, Class I, Type A shock requirements of MIL-S-901.

4.5.9 Vibration resistance. RAO universal joints installed between valves and actuators shall be tested in accordance with the Type I requirements of MIL-STD-167-1.

4.5.10 Detrimental materials.

4.5.10.1 Heavy metals. Heavy metals shall be analyzed in accordance with the methods specified in ASTM D3223, D3557, D3559, or D4190 (see 3.2.1), or by review of material certifications of analysis.

4.6 Retention of qualification. To retain qualification, the contractor shall submit a test report to the qualifying activity at 24 month intervals. The qualifying activity shall establish the initial reporting date.

Each report shall consist of a summary of test and inspection results required by this specification that were performed during the 24 month reporting interval. As a minimum, the report shall include the following:

a. Number of lots produced and tested, including lot and sample sizes for each lot.

b. Identify which tests were performed.

c. Quantities passed.

d. Quantities failed.

e. All reworked sampling lots shall be accounted for and identified. A summary of corrective action taken shall be included.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. RAO universal joints are designed to transmit up to 18,000 inch-pounds of torque between misaligned and/or offset shafts. RAO universal (double) joints accommodate compound angles up to 90 degrees (45 degrees each joint) replacing pin-and-cube universal joints and gear boxes in remote operator arrangements installed on Navy surface ships and submarines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity required.
c. Part or identifying number (PIN) (see 1.3).
d. Applicable National Stock Number (NSN) if available.
e. Specific issue of individual documents referenced (see 2.2.1 and 2.3).
f. Marking requirements (see 3.5.2).
g. Packaging requirements (see 5.1).
h. Shelf-life requirements (see 6.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 32232 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil.

6.4 Shelf-life. This specification covers items where shelf-life is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order. The shelf-life codes are contained in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M; Shelf-Life Management Manual, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points (ICPs), and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: http://www.shelflife.hq.dla.mil/.

6.5 Subject term (key word) listing.

Gear box

Pin-and-cube

Remote operating gear

ROG

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Air Force - 99

Preparing Activity:
Navy - SH
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Review Activities:
Air Force - 71
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